Date

DAY - 11 (After Mid 2)

FOURIER SERIES:- (Ch#6)

Signal can be represented as a sum of its components in a variety of ways.

$$g(t) \cong c_1 x_1(t) + c_2 x_2(t) + \cdots + c_N x_N(t)$$
$$= \sum_{n=1}^{N} c_n x_n(t) \qquad t_1 \le t \le t_2$$

the Parseval's theorem says that the energy of the sum of orthogonal signal components is equal to the sum of the signal's energy.

Orthogonal Basis:-

Just as vector cordinates systems can be formed by mutually orthogonal vectors (e.g. rectangular, cylindrical, spherical), we can have signal cordinate system (basis signals) formed by a variety of mutually orthogonal signals! By definition

Orthogonal signals $\int_{t_1}^{t_2} g(t) \cdot x(t)\, dt = 0$

Two complex functions $x_1(t)$ and $x_2(t)$ are orthogonal over an interval $(t_1 \le t \le t_2)$

$$\int_{t_1}^{t_2} x_1(t)\, x_2^*(t)\, dt = 0 \quad \text{or} \quad \int_{t_1}^{t_2} x_1^*(t)\, x_2(t)\, dt = 0$$

we can define an orthogonal signal space as

$$\int_{t_1}^{t_2} x_m(t)\, x_n^*(t)\, dt = \begin{cases} 0 & m \ne n \\ E_n & m = n \end{cases}$$

Every signal $g(t)$ can be approximated over an

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interval $[t_1, t_2]$ by a set of N mutually orthogonal signals -

$x_1(t), x_2(t), x_3(t) \cdots x_N(t)$

there exists large number of orthogonal signals which can be used as basis signals for the generalized fourier series e.g. sinusoid functions, exponential functions walsh functions, Bessel functions, etc.

A periodic signal g(t) can be expressed as a sum of sinusoids of frequencies 0 (dc), w_0, $2w_0$ ...., nw_0 whose amplitudes are $C_0, C_1, C_2, \ldots C_n$.

A famous set of orthogonal signals are the sinusoidal signals and

$$g(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nw_0 t + b_n \sin nw_0 t)$$

$t_1 \leq t \leq t_2$

where $w_0 = \frac{2\pi}{T_0}$

The Fourier Coefficients :-

a_0, a_n, and b_n are

$$a_0 = \frac{1}{T_0} \int_{t_1}^{t_1+T_0} g(t)\, dt$$

$$a_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} g(t) \cos nw_0 t\, dt$$

$$b_n = \frac{2}{T_0} \int_{t_1}^{t_1+T_0} g(t) \sin nw_0 t\, dt$$

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The fourier series can be plotted as amplitude spectrum [C_n vs ω] or the phase spectrum [θ_n vs ω] which form the frequency spectra of $g(t)$.

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FOURIER SERIES:

$$g(t) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t\right)$$

A signal has a dual identity time domain identity $\phi(t)$ and the frequency domain identity [Fourier spectra].
Two identities complement each other.

Existance Of Fourier Series (Dirichlet conditions)

1. For the series to exist, a_0, a_n and b_n must be finite

Absolute Integrability

$$\int_{T_0} |g(t)|\, dt < \infty$$

weak Dirichlet conditions, Fourier series is guaranteed to exist but may not converge.

2. $g(t)$ has only finite no. of maxima and minima in one period and it may have only finite discontinuities in one period.
Combined with D → Form Strong Dirichlet conditions

→ All laboratory signals satisfy both the conditions and have convergent fourier series.

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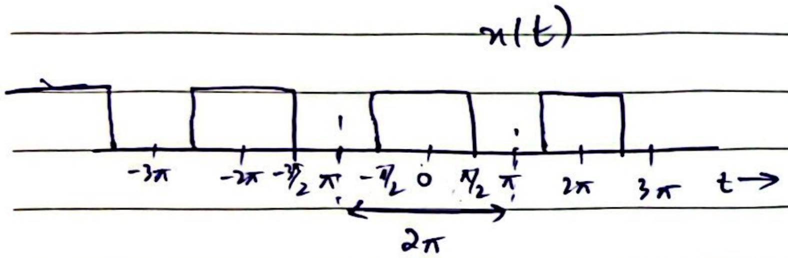
Fourier representation is a way of expressing a signal in terms of ever lasting sinusoids.

Balance of magnitude and phases of components of the fourier spectrum add up to precisely produce the pulse g(t).

Question :-



Find the fourier series for the square-pulse periodic signal

$T_0 = 2\pi$ and $\omega_0 = \frac{2\pi}{T_0} = 1$

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt) \quad [\omega_0 = 1]$$

$$a_0 = \frac{1}{T_0}\int_{T_0} x(t)\,dt = \frac{1}{2\pi}\int_{-\pi/2}^{\pi/2} 1\,dt \Rightarrow \frac{1}{2}$$

$$a_n = \frac{1}{\pi} = \int_{-\pi/2}^{\pi/2} \cos nt\,dt = \frac{2}{n\pi}\sin\left(\frac{n\pi}{2}\right)$$

$$a_n = \begin{cases} 0 & n \text{ even} \\ \frac{2}{\pi n} & n = 1, 5, 9, 13, \ldots \\ \frac{-2}{n\pi} & n = 3, 7, 11, 15 \ldots \end{cases}$$

$$b_n = \frac{1}{\pi}\int_{-\pi/2}^{\pi/2} \sin nt\,dt \Rightarrow 0$$

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A. $x(t) = \frac{1}{2} + \frac{2}{\pi}\left(\cos t - \frac{1}{3}\cos 3t + \frac{1}{5}\cos 5t - \frac{1}{7}\cos 7t + \cdots\right)$

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A. $x(t) = \frac{1}{2} + \frac{2}{\pi}\left(\cos t - \frac{1}{3}\cos 3t + \frac{1}{5}\cos 5t - \frac{1}{7}\cos 7t + \cdots\right)$

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Compact Trigonometric Fourier Series :-

When $x(t)$ is real, cofficients a_n and b_n are real for all n, and the trigonometric fourier series can be expressed in compact form

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

where C_n and θ_n are related to a_n and b_n as

$$C_0 = a_0$$

$$C_n = \sqrt{a_n^2 + b_n^2}$$

$$\theta_n = \tan^{-1}\left(\frac{-b_n}{a_n}\right)$$

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Periodic Signal :-

A signal $f(t)$ is said to be periodic if for some positive constant T_0

$$f(t) = f(t + T_0) \quad \text{for all } t \rightarrow ①$$

the smallest value of T_0 which satisfies eq ① is called the period of $f(t)$.

A periodic signal $f(t)$

1. must be everlasting
2. $f(t)$ can be generated by periodic extension of any segment of any segment of $f(t)$ of duration T_0 (the period)

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3. Area under $f(t)$ over any interval of duration T_0 is same.

$f(t)$ can be expressed as

$$f(t) = C_0 + C_1 \cos(\omega_0 t + \theta_1) + C_2 \cos(\omega_0 t + \theta_2) + \dots$$

For generality, all harmonics upto infinity will be included.

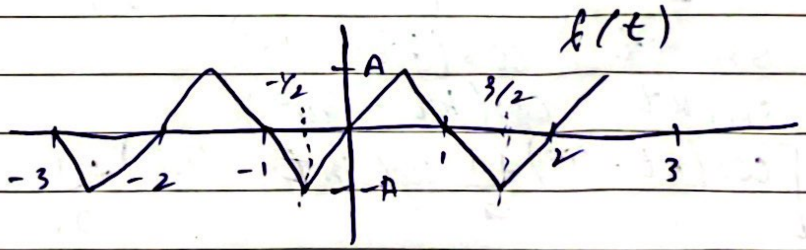
$$f(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

$$f(t+T_0) = C_0 + \sum_{n=1}^{\infty} C_n \cos[n\omega_0(t+T_0) + \theta_n]$$

$$f(t+T_0) = C_0 + \sum_{n=1}^{\infty} C_n \cos[(n\omega_0 t + \cancel{n\omega_0 T_0} \cdot 2\pi n) + \theta_n]$$

$$= C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n) \Rightarrow f(t)$$

the fourier series for any even periodic function $x(t)$ consist of cosine terms only and the series for any odd periodic function $x(t)$ consist of sine terms only.



Find the compact trigonometric fourier series for the triangular periodic signal $f(t)$ shown sketch the amplitude and phase spectra of $f(t)$.

$T_0 = 2$ $\qquad \omega_0 = \frac{2\pi}{T_0} \Rightarrow \pi$

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$$f(t) = a_0 + \sum_{n=1}^{\infty} \left\{ a_n \cos n\pi t + b_n \sin n\pi t \right\}$$

where $f(t) = \begin{cases} 2At & |t| < 1/2 \\ 2A(1-t) & 1/2 < t \le 3/2 \end{cases}$

Average value (dc) of $f(t)$ is zero $\therefore$ $\boxed{a_0 = 0}$

$$a_n = \frac{2}{2} \int_{-1/2}^{3/2} f(t) \cos n\pi t \, dt$$

$$= \int_{-1/2}^{3/2} 2At \cos n\pi t \, dt + \int_{1/2}^{3/2} 2A(1-t) \cos n\pi t \, dt$$

$$\boxed{a_n = 0}$$

$$b_n = \int_{-1/2}^{1/2} 2At \sin n\pi t \, dt + \int_{1/2}^{3/2} 2A(1-t) \sin \pi t \, dt$$

$$\boxed{b_n = \frac{8A}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right)}$$

$$f(t) = \frac{8A}{\pi^2} \sum_{n=1,3,5..}^{\infty} \left\{ \frac{\sin(n\pi/2)}{n^2} \sin n\pi t \right\}$$

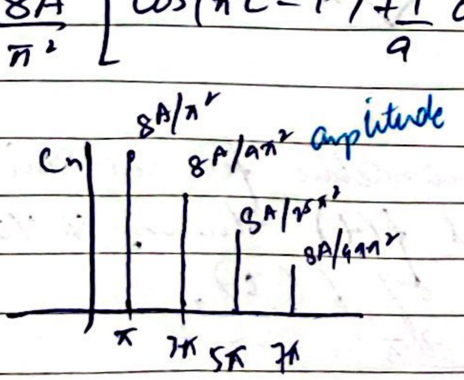
$$f(t) = \frac{8A}{\pi^2} \left[\sin \pi t - \frac{1}{9} \sin 3\pi t + \frac{1}{25} \sin 5\pi t - \frac{1}{49} \sin 7\pi t - \right.$$

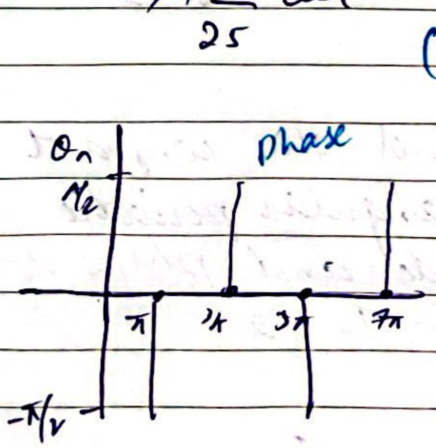
Identities $\sin kt = \cos(kt - 90^\circ)$

$-\sin kt = \cos(kt + 90^\circ)$

$$f(t) = \frac{8A}{\pi^2} \left[\cos(\pi t - 90^\circ) + \frac{1}{9} \cos(3\pi t + 90^\circ) + \frac{1}{25} \cos(5\pi t - 90^\circ) \ldots \right]$$

(compact form)





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The Fourier Spectrum :-

The compact trigonometric fourier series

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n) \rightarrow (A)$$

indicates that a periodic signal $x(t)$ can be expressed as a sum of sinusoids of frequencies 0 (dc), ω_0, $2\omega_0$, ... $n\omega_0$, whose phases are $0, \theta, \theta_2, \theta_3, \ldots \theta_n$ respectively.

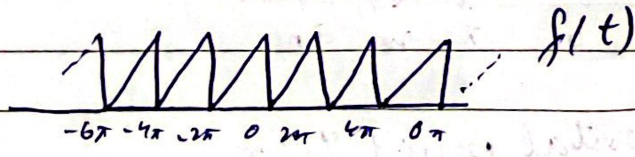
We can readily plot C_n verses n (phase spectrum). These plots are the frequency spectra of $x(t)$.

Knowing these spectra, we can reconstruct or synthesise the signal $x(t)$ according to equation (A).

The frequency spectra of a signal constitutes a frequency domain description of $x(t)$. In contrast to the time domain description, where $x(t)$ is specified as a function of time.

Question

Find the compact trigonometric fourier series and sketch the amplitude and phase spectra of $f(t)$.



solution :-

$T_0 = 2\pi$ $\qquad \omega_0 = \frac{2\pi}{T_0} \Rightarrow 1$

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt)$$

DAY-14

$a_0 = 0.5$ (by inspection)

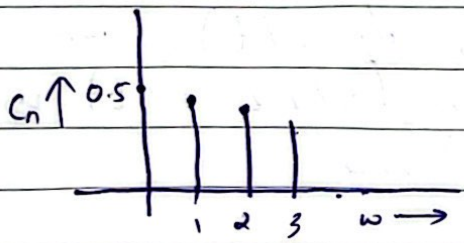
$$a_n = \frac{2}{T_0}\int_{t_1}^{t_1+T} f(t)\cos nt\,dt$$

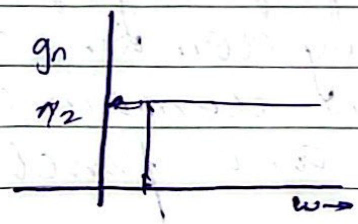
$$a_n = \frac{1}{\pi}\int_0^{2\pi} \frac{t}{2\pi}\cos nt\,dt = 0$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} \frac{t}{2\pi}\sin nt\,dt = \frac{1}{\pi n}$$

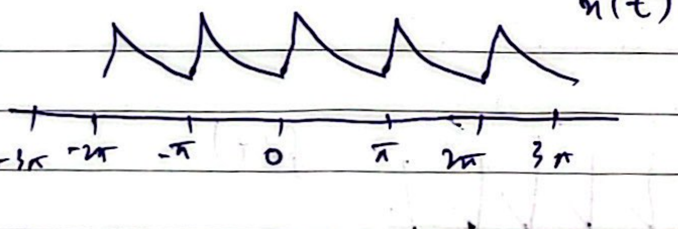
$$f(t) = 0.5 - \frac{1}{\pi}\left(\sin t + \frac{1}{2}\sin 2t + \frac{1}{3}\sin t \cdots\right)$$

$$f(t) = 0.5 + \frac{1}{\pi}\left[\cos\left(t+\frac{\pi}{2}\right) + \frac{1}{2}\cos\left(2t+\frac{\pi}{2}\right) + \frac{1}{3}\cos\left(5t+\frac{\pi}{2}\right) + \cdots\right]$$





Question :-



the fundamental period $T_0 = \pi$

$$f_0 = \frac{1}{T_0} = \frac{1}{\pi}, \quad \omega_0 = \frac{2\pi}{T_0} = 2\pi f = 2\pi \cdot \frac{1}{\pi} \Rightarrow 2$$

$$x(t) = a_0 + \sum_{n=1}^{\infty}\left(a_n\cos 2nt + b_n\sin 2nt\right)$$

$$a_0 = \frac{1}{\pi}\int_{T_0} x(t)\,dt \Rightarrow a_0 = \frac{1}{\pi}\int_0^{\pi} e^{-t/2}\,dt = 0$$

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$$a_n = \frac{2}{\pi}\int_0^{\pi} e^{-t/2}\cos 2nt \, dt = 0.504\left(\frac{2}{1+16n^2}\right)$$

$$b_n = \frac{2}{\pi}\int_0^{\pi} e^{-t/2}\sin 2nt \, dt = 0.504\left(\frac{8n}{1+16n^2}\right)$$

$$x(t) = 0.504\left[1 + \sum_{n=1}^{\infty} \frac{2}{1+16n^2}\left(\cos 2nt + 4n \sin 2nt\right)\right]$$

$$C_0 = a_0 = 0.504$$

$$C_n = \sqrt{a_n^2 + b_n^2} = 0.504\sqrt{\frac{4}{(1+16n^2)^2} + \frac{64n^2}{(1+16n^2)^2}}$$

$$= 0.504\left(\frac{2}{\sqrt{1+16n^2}}\right)$$

$$\theta_n = \tan^{-1}\left(\frac{-b_n}{a_n}\right) = \tan^{-1}(-4n) = -\tan^{-1} 4n$$

$$x(t) = 0.504 + 0.504\sum_{n=1}^{\infty} \frac{2}{\sqrt{1+16n^2}}\cos\left(2nt - \tan^{-1} 4n\right)$$

DAY-15

Exponential Fourier Series :-

Euler's identity

$e^{j\pi} + 1 = 0$ which leads to

$e^{j\omega t} = \cos\omega t + j\sin\omega t \rightarrow$ A

A periodic signal $x(t)$ can be expressed as

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \rightarrow 1$$

Multiply 1 with $e^{-jm\omega_0 t}$ and integrate over one period

$$\int_{T_0} x(t)e^{-jm\omega_0 t}\,dt = \sum_{n=-\infty}^{\infty} D_n \int_{T_0} e^{j(n-m)\omega_0 t}\,dt \rightarrow 2$$

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Orthogonality propety :-

It says

$$\int_{T_0} e^{jm\omega_0 t} e^{-jn\omega_0 t} dt = \begin{cases} 0 & m \neq n \\ T_0 & m = n \end{cases} \rightarrow (3)$$

Use (3) in (2)

$$D_n T_0 = \int_{T_0} x(t) e^{-jm\omega_0 t} dt \rightarrow (4)$$

$$\boxed{D_n = \frac{1}{T_0} \int_{T_0} x(t) e^{-jm\omega_0 t} dt}$$

To summarize the exponential fourier series

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$

$$D_n = \frac{1}{T_0} \int_{T_0} x(t) e^{-jm\omega_0 t} dt$$

Fourier Series representation of periodic signal of period T_0 $\left(\omega_0 = \frac{2\pi}{T_0}\right)$

Series Form

DAY-12

Trigonometric

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Compact trigonometric

$$f(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_0)$$

Exponential

$$f(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$

Coefficient computation

$$a_0 = \frac{1}{T_0} \int_{T_0} f(t) dt$$

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$$a_n = \frac{2}{T_0} \int_{T_0} f(t) \cos n\omega_0 t \, dt$$

$$b_n = \frac{2}{T_0} \int_{T_0} f(t) \sin n\omega_0 t \, dt$$

$$C_0 = a_0$$

$$C_n = \sqrt{a_n^2 + b_n^2}$$

$$\theta_n = \tan^{-1}\left(\frac{-b_n}{a_n}\right)$$

$$D_n = \frac{1}{T_0} \int_{T_0} f(t) \, e^{-jn\omega_0 t} \, dt$$

Conversion Formulas

$$a_0 = C_0 = D_0$$

$$a_n - j b_n = C_n e^{j\theta_n} = 2D_n$$

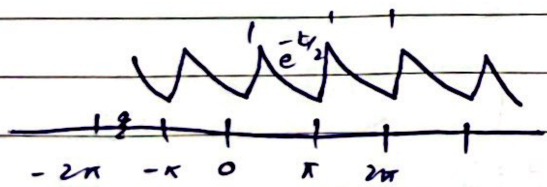
$$a_n + j b_n = C_n e^{-j\theta_n} = 2D_{-n}$$

$$C_0 = D_0$$

$$C_n = 2|D_n| \qquad : n \geq 1$$

$$\theta_n = \angle D_n$$

QUESTION :-



Find the exponential fourier series for $x(t)$

sol.

$$T_0 = \pi \qquad \omega_0 = \frac{2\pi}{T_0} \Rightarrow 2$$

Quiz on next Monday on trigonometric

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$$x(t) = \sum_{n=-\infty}^{\infty} D_n \, e^{j2nt}$$

where

$$D_n = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-j2nt}\, dt = \frac{1}{\pi}\int_0^{\pi} e^{-t/2} e^{-j2nt}\, dt$$

$$= \frac{1}{\pi}\int_0^{\pi} e^{-(1/2+j2n)t}\, dt = \frac{-1}{\pi(1/2+2jn)} e^{-(1/2+j2n)t}\Big|_0^{\pi}$$

$$D_n = \frac{0.504}{1+j4n}$$

$$x(t) = 0.504 \sum_{n=-\infty}^{\infty} \frac{1}{1+j4n} e^{j2nt}$$

$$= 0.504\left[1 + \frac{1}{1+j4} e^{j2t} + \frac{1}{1+j8} e^{j4t} + \dots + \frac{1}{1-j4} e^{-j2t} + \frac{1}{1-j8} e^{-j4t} + \dots\right]$$

DAY-16

Exponential Fourier Series :-

$$x(t) = \sum_{n=-\infty}^{\infty} D_n \, e^{jn\omega_0 t}$$ — Generic exponential fourier series.

$$x(t) = 0.504 \sum_{n=-\infty}^{\infty} \frac{1}{1+j4n} e^{j2nt}$$

$$= 0.504\left[1 + \frac{1}{1+j4} e^{j2t} + \frac{1}{1+j8} e^{j4t} + \dots + \frac{1}{1-j4} e^{-j2t} + \frac{1}{1-j8} e^{-j4t} + \dots\right]$$

Observe that the coefficients D_n are complex. Moreover, D_n and D_{-n} are conjugate.

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For the exponential fourier series, we plot the magnitude $|D_n|$ and the angle of D_n $\angle D_n$

this requires that the coefficients D_n be expressed in polar form as

$$|D_n| \, e^{j\angle D_n}$$

For the above series

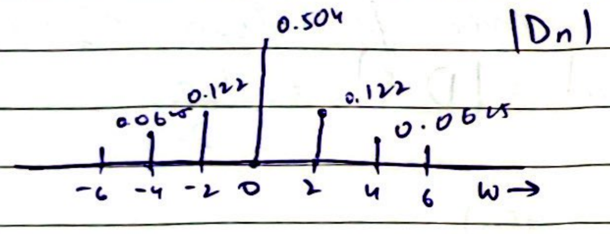
$$D_0 = 0.504$$

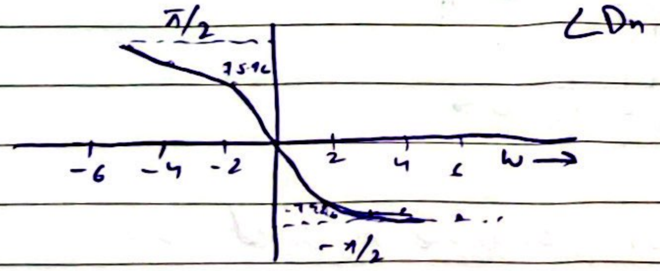
$$D_1 = \frac{0.504}{1+j4} = 0.122\, e^{-j75.96} \Rightarrow |D_1| = 0.122$$
$$\angle D_1 = -75.96°$$

$$D_{-1} = \frac{0.504}{1-j4} = 0.122\, e^{j75.96°} \Rightarrow |D_{-1}| = 0.122$$
$$\angle D_{-1} = 75.96°$$

$$D_2 = \frac{0.504}{1+j8} = 0.0625\, e^{-j82.87°} \Rightarrow |D_2| = 0.0625$$
$$\angle D_2 = -82.87°$$

$$D_{-2} = \frac{0.504}{1-j8} = 0.0625\, e^{j82.87} \Rightarrow |D_{-2}| = 0.0625$$
$$\angle D_{-2} = 82.87°$$





QUESTION :-

The trigonometric fourier series is shown, sketch the

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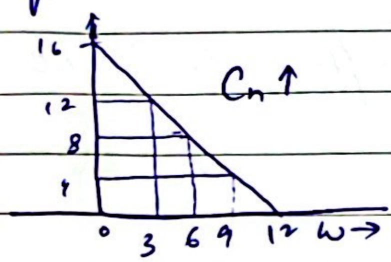
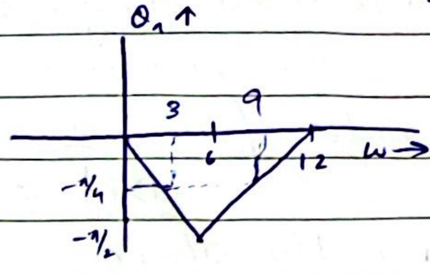
exponential fourier series and verify result analytically.





The trigonometric fourier series has four spectral components at ω freq. 0, 3, 6 and 9.

the DC component is 16.

Phase component of ω freq. 3 and 9 is $\left(-\frac{\pi}{4}\right)$

Phase component of ω freq. 6 is $-\frac{\pi}{2}$

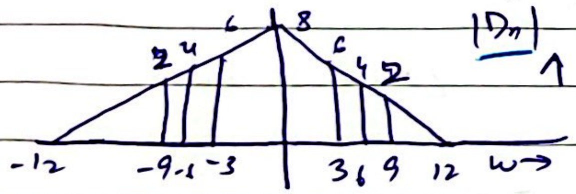
$$x(t) = 16 + 12\cos(3t - \pi/4) + 8\cos(6t - \pi/2) + 4\cos(9t - \pi/4)$$

For the exponential fourier series

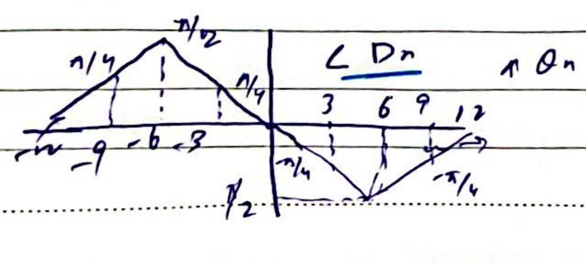
$$D_0 = C_0 = 16$$

$|D_n|$ is an even function of ω

$$|D_n| = |D_{-n}| = \frac{C_n}{2}$$



$\angle D_n = \theta_n$ for positive n and $-\theta_n$ for negative n



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$$\hat{x}(t) = 16 + \left[6e^{-j\pi/4}e^{j3t} + 6e^{j\pi/4}e^{-j3t}\right] + \left[4e^{-j\pi/2}e^{j6t} + 4e^{j\pi/2}e^{-j6t}\right] + \left[2e^{-j\pi/4}e^{j9t} + 2e^{j\pi/4}e^{-j9t}\right]$$

$$\hat{x}(t) = 16 + 6\left[e^{j(3t-\pi/4)} + e^{-j(3t-\pi/4)}\right] + 4\left[e^{j(6t-\pi/2)} + e^{-j(6t-\pi/2)}\right] + 2\left[e^{j(9t-\pi/4)} + e^{-j(9t-\pi/4)}\right]$$

$$\boxed{x(t) = 16 + 12\cos(3t - \pi/4) + 8\cos(6t - \pi/2) + 4\cos(9t - \pi/4)}$$

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$$\hat{x}(t) = 16 + \left[6 e^{-j\pi/4} e^{j3t} + 6 e^{j\pi/4} e^{-j3t}\right] + \left[4 e^{-j\pi/2} e^{j6t} + 4 e^{j\pi/2} e^{-j6t}\right] + \left[2 e^{-j\pi/4} e^{j9t} + 2 e^{j\pi/4} e^{-j9t}\right]$$

$$\hat{x}(t) = 16 + 6\left[e^{j(3t-\pi/4)} + e^{-j(3t-\pi/4)}\right] + 4\left[e^{j(6t-\pi/2)} + e^{-j(6t-\pi/2)}\right] + 2\left[e^{j(9t-\pi/4)} + e^{-j(9t-\pi/4)}\right]$$

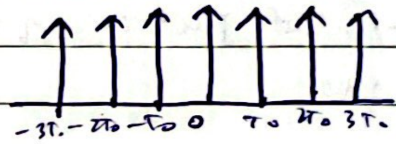
$$\boxed{x(t) = 16 + 12\cos(3t - \pi/4) + 8\cos(6t - \pi/2) + 4\cos(9t - \pi/4)}$$

DAY- 17

Impulse train and its fourier series :-

Impulse train is the key for analogue -to- digital conversion of signals.

$x(t) = \delta_{T_0}(t)$



the unit impulse train can be expressed as

$$\delta_{T_0}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$$

the exponential fourier series for this impulse train would be

$$\delta_{T_0}(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \qquad \omega_0 = \frac{2\pi}{T_0}$$

where

$$D_n = \frac{1}{T_0}\int_{T_0} \delta_{T_0}(t)\, e^{-jn\omega_0 t}\, dt$$

over the interval $\left(\frac{-T_0}{2}, \frac{T_0}{2}\right) \rightarrow \delta_{T_0}(t) = \delta(t)$

$$\left[\int_{-\infty}^{\infty} \delta(t)\,dt = 1\right]$$

Impulse is located at $t = 0$, so using

$$\int_{-\infty}^{\infty} \phi(t)\delta(t)\,dt \qquad \text{[sampling property]}$$

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$$= \phi(0)\int_{-\infty}^{\infty} \delta(t)\, dt$$

$$= \phi(0)$$

So

$$\int_{-T/2}^{T/2} \delta(t)\, e^{-jn\omega_0 t}\, dt$$

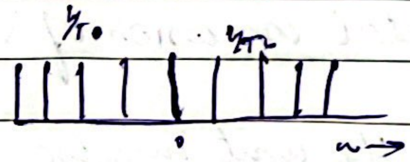
is evaluated at $t=0$ and becomes 1.

So $D_n = \frac{1}{T_0}$ FI-VACT

Now the exponential fourier series is

$$\delta_{T_0}(t) = \frac{1}{T_0} \sum_{n=-\infty}^{\infty} e^{jn\omega_0 t}$$

is the exponential fourier series



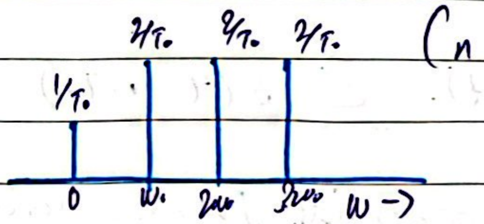
Only amplitude plot is required as the phases are zero.

Now,

$$C_0 = D_0 = 1/T_0$$

$$C_n = 2|D_n| = 2/T_0.$$

$$\theta_n = 0.$$



and

$$\delta_{T_0}(t) = \frac{1}{T_0}\left[1 + 2(\cos\omega_0 t + \cos 2\omega_0 t + \cos 3\omega_0 t + \dots)\right] \qquad \omega_0 = \frac{2\pi}{T_0}$$

is the trigonometric Fourier Series for impulse train $\delta_{T_0}(t)$.

DAY-16

Fourier Transform :-

By applying the limiting process, an aperiodic

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signal can be expressed as a continuous sum (integral) of everlasting exponentials.

$$x(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\omega)\, e^{j\omega t}\, d\omega$$

is known as the Fourier integral.

$X(\omega) = F[x(t)]$ is the direct fourier transform of $x(t)$.

whereas

$x(t) = F^{-1}[X(\omega)]$ is the inverse fourier transform of $X(\omega)$

$x(t) \Leftrightarrow X(\omega)$

$$\boxed{\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt \\ x(t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega \end{aligned}}$$

the transform $X(\omega)$ is the frequency domain specification of $x(t)$

the spectrum $X(\omega)$ can be plotted as a fraction of ω

$X(\omega) = |X(\omega)| e^{j(X(\omega))}$

$X(\omega)$ which is the fourier transform of $x(t)$ is more precisely the spectral density per unit bandwidth

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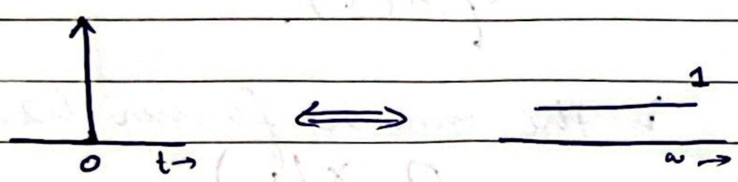
Question 1:-

Find the fourier transform of unit impulse function $\delta(t)$

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

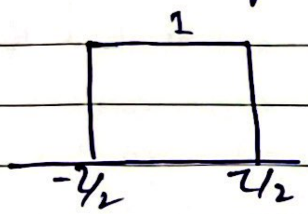
$$X(\omega) = \int_{-\infty}^{\infty} \delta(t)\, e^{-j\omega t}\, dt$$

$$= 1$$



Question 2:-

Find the fourier transform of $g(t) = \{ \text{rect}\left(\frac{t}{\tau}\right)^2 \}$.



$$g(t) \Longleftrightarrow G(\omega)$$

$$G(\omega) = ??$$

we have $G(\omega) = \int_{-\infty}^{\infty} g(t)\, e^{-j\omega t}\, dt = \int_{-\infty}^{\infty} \text{rect}\left(\frac{t}{\tau}\right) e^{-j\omega t}\, dt$

since $\text{rect}\left(\frac{t}{\tau}\right) = 1$ for $|t| \le \frac{\tau}{2}$ and

since $\text{rect}\left(\frac{t}{\tau}\right) = 0$ for $|t| > \frac{\tau}{2}$

$$G(\omega) = \int_{\tau/2}^{\tau/2} 1 \cdot e^{-j\omega t}\, dt = \frac{-1}{j\omega}\left(e^{-j\omega \tau/2} - e^{j\omega \tau/2}\right)$$

$$G(\omega) = \frac{2 \sin \omega \tau/2}{\omega}$$

$$G(\omega) = \frac{2 \sin(\omega\tau/2)}{\omega}$$

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$$G(\omega) = \tau \frac{\sin\left(\frac{\omega\tau}{2}\right)}{\left(\omega\tau/2\right)}$$

$$= \tau \operatorname{sinc}\left(\frac{\omega\tau}{2}\right)$$

$$\operatorname{rect}\left(\frac{t}{\tau}\right) \Longleftrightarrow \tau \operatorname{sinc}\left(\frac{\omega\tau}{2}\right)$$